Tutorial - 6

In this tutorial, you will develop and test collective communication function calls in MPI, compute a row block-decomposed matrix-vector multiplication, and also test MPI_Type_create_struct function.

1. Develop MPI programs to demonstrate collective functions such as MPI_Reduce, MPI_Allreduce, MPI_Scatter, MPI_Gather, MPI_Allgather. Convince yourself that these programs are producing the intended output by running them.

2. Develop an MPI program that computes matrix-vector multiplication and produces the resulting vector using block-decomposition along rows. Convince yourself that you obtain the same answer as that of the serial program. Use collective communication functions as much as possible. Run the program on $p = 2, 4$ and 8 processes.

3. Consider the following variable declarations, the values assigned are for indicative purpose. Write an MPI program that creates a derived data type using MPI_Type_create_struct function and broadcast these values to other processes in one go. Test your program with multiple processes.

```
1   double a[3] = {12.67, 56.45, 25.32};
2   double b[2] = {56.79, 98.26};
3   int n = 10;
4
```